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EDUCATION

2026–2027 **University of Wisconsin–Madison**
M.S. in Physics: Quantum Computing (MSPQC). Expected May 2027.
Relevant coursework: Quantum Computing Laboratory, Qubit Tune-Up and Programming, Solid State, Graduate Seminar on Quantum Computing.

2022–2026 **University of Wisconsin–Madison**
B.S. in Computer Science and Mathematics; Physics Certificate.
Relevant coursework: Intro to Quantum Computing, Systems Architecture for Quantum Computing, Quantum Mechanics, Algorithms, Real Analysis, Abstract Algebra, Combinatorics, Operating Systems, Linear Optimization, Theory of Programming Languages.

RESEARCH

2025– **Distributed Quantum Computing**, Argonne National Laboratory.
Advisor: Dr. Rajkumar Kettimuthu.
• Designed and built **SeQUeNCe.DQC**, a top-level module turning SeQUeNCe (2026 R&D 100 Winner) into a full-stack DQC simulator: circuit front end, compiler, central controller, and per-node worker agents.
• Implemented DQC nodes with separate data and communication memory, TeleGate/TeleData teleportation primitives.
• Built QAP, random, and time-sliced FGP-rOEE compilers behind a compiler-agnostic interface, plus ket-vector trajectory and density-matrix CPTP simulation backends.
• Modeled the classical controller as a real network node, and added a trajectory noise model with per-qubit idle T1/T2 decoherence and per-shot Pauli errors.
• Reproduced the 71% distributed Grover success rate of a *Nature* trapped-ion experiment under both lumped and full per-source error budgets.

2024– **Fault Tolerance & Benchmarking**, University of Wisconsin–Madison.
Advisor: Prof. Swamit Tannu.
• Currently developing compilation strategies for bivariate bicycle codes.
• Developed benchmark suites for fault-tolerant quantum hardware; used variational autoencoders to learn latent-space representations of quantum circuits, enabling similarity analysis based on metrics such as T-gate resource estimation.

Apr 2026– **Graph State Verification**, University of Oxford (remote).
Collaborator: Dr. Sathyawageeswar Subramanian.
• Extending graph state verification protocols to more sophisticated Pauli noise models for fault-tolerant measurement-based quantum computing.

2023– **Quantum Algorithms for Scientific Computing**, University of Wisconsin–Madison.
Advisor: Prof. Krishnan Suresh.
• Formulated sparse approximate inverse (SPAI) preconditioner construction as a QUBO and computed it on D-Wave Advantage hardware to accelerate conjugate gradient solves of poorly conditioned linear systems.
• Reduced the box-algorithm contraction ratio for 30–40% faster solution of random

linear systems; adapted quantum annealing to inequality-constrained portfolio optimization.
• Developed efficient encoding schemes for structured quantum states (PyEncode), with applications to engineering simulation.

Publications

2027 **SeQUeNCe.DQC: Full-stack Simulation of Distributed Quantum Computing.** Sanjay Suresh, Caitao Zhan, Joaquin Chung, Rajkumar Kettimuthu. *IEEE International Conference on Communications*, under review.

2026 **PyEncode: An Open-Source Library for Structured Quantum State Preparation.** Krishnan Suresh, Sanjay Suresh. *Quantum Science & Technology*.

2025 **Computing a Sparse Approximate Inverse on Quantum Annealing Machines.** Sanjay Suresh, Krishnan Suresh. *Journal of Computing and Information Science in Engineering*.

2024 **Optimal Box Contraction for Solving Linear Systems via Simulated and Quantum Annealing.** Sanjay Suresh, Krishnan Suresh. *Engineering Optimization*.

2021 **A Visual Approach to Programming for Middle and High School Students.** Sanjay Suresh. ISBN 979-8470081841.

Talks & Posters

2026 Workshop co-instructor, *Quantum Computing for Engineers*, QC 2026; poster on PyEncode.

2025 Poster on distributed quantum computing with SeQUeNCe, Chicago Quantum Summit.

2024 Talk and poster on quantum annealing for linear systems and constrained optimization, WCCM, Vancouver.

Projects

2025– **SurfaceScope**, an interactive tool to teach quantum error correction (in progress).
• Visualizes Stim-sampled errors, syndrome extraction, and decoding on a surface code, step by step; expanding toward support for more codes.
Python, Flask, React, TypeScript, Stim, PyMatching.

2025–2026 **Studia**, AI tutoring platform.
• Full-stack platform with structured lessons, PDF context extraction via RAG, automated review generation, and real-time Manim animations. Deployed in CS 639 (Prof. Swamit Tannu).
Next.js, TypeScript, Convex, Vercel AI SDK, Manim.

Teaching & Outreach

2019– **Principal Instructor**, Printable Programming, Madison, WI.
• Authored a CAD modelling, programming and 3D-printing curriculum; taught STEM at the Boys and Girls Club of Dane County; contributed to the CADjs software application.

2020–2022 **President & Captain**, Science Bowl, Middleton, WI.
• Founded Summit Scibowl for advanced student training; built online platforms engaging 2,000+ students nationwide and organized national tournaments.

Honors & Awards

2022 Atlas Fellowship — selected from 7,000+ applicants worldwide (114 recipients) for a $50K fellowship.

2022 Wisconsin Math Talent Search Finalist; Mathematics Student of the Year, Middleton High School.

2021, 2022 Chemistry Olympiad National Qualifier; National Astrophysics Olympiad Qualifier (2021); AIME Qualifier.

2020 Naval Research Award, Capital Science and Engineering Fair; Prometheus Science Bowl, 2nd nationally.

Skills

Quantum Qiskit, Stim, PyMatching, PsiQuantum Workbench, SeQUeNCe, pytket / pytket-dqc, PennyLane, QuTiP, D-Wave Ocean.

Languages Python, TypeScript, C, Java, Haskell, MATLAB.

Web & Infra Next.js, React, Convex, Vercel AI SDK, Flask, PostgreSQL, Docker, Git.

Other

Music Carnatic violin. Professional violinist and accompanist to frontline artists at major festivals across India and the US. Trained since 2012; performing as soloist and accompanist since 2017.